

CADDIE: Compact Adaptive Detection-Driven Inference for Real-Time Golf Club Pose Estimation

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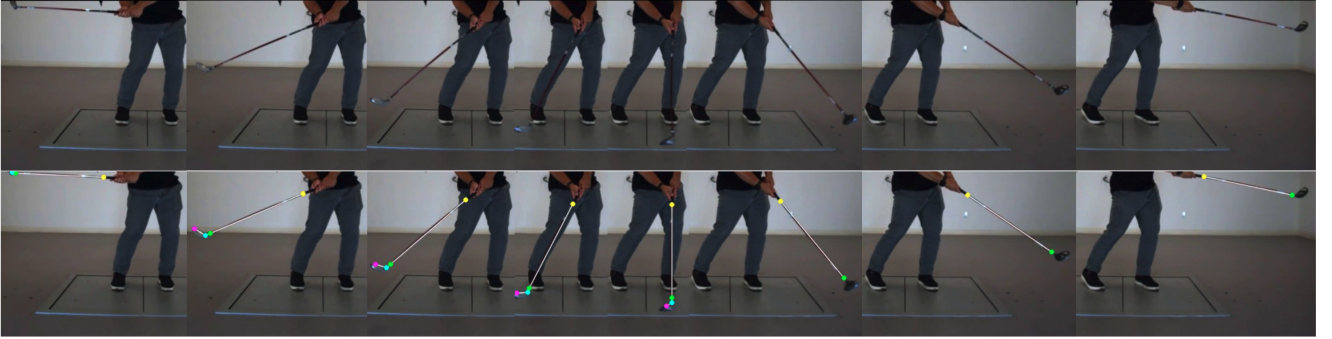


Figure 1. Golf Club Keypoints in Swing Motion.

Abstract

methods in both keypoint accuracy and inference speed.

Golf performance depends on precise coordination between players and their clubs. Most vision systems focus on body pose rather than club motion, which is key to analyzing swing speed, trajectory, and carry-at-impact. We address this gap with CADDIE, a compact markerless two-stage framework tailored to the unique challenges of golf club pose estimation including thin elongated geometry, motion blur, and self-occlusion. We introduce Golf-Club, a dataset of 72K annotated frames across 7 subjects, 6 locations, and 3 club types including synchronized multiview sequences. The dataset is paired with a universal five-keypoint schema covering the shaft and club head that applies consistently to drivers, woods, and irons. CADDIE pairs lightweight ConvNeXtV2-Atto backbones (14.3M total parameters) with RetinaNet detection and heatmap-based keypoint regression. An adaptive spatial-temporal smoothing pipeline further scales throughput from 69 to 238 FPS on a single RTX 3090 with less than 1% accuracy degradation. CADDIE achieves 88.52% PCK@5px and outperforms HRNet (28.6M), ViTPose (90.0M), and YOLO-based

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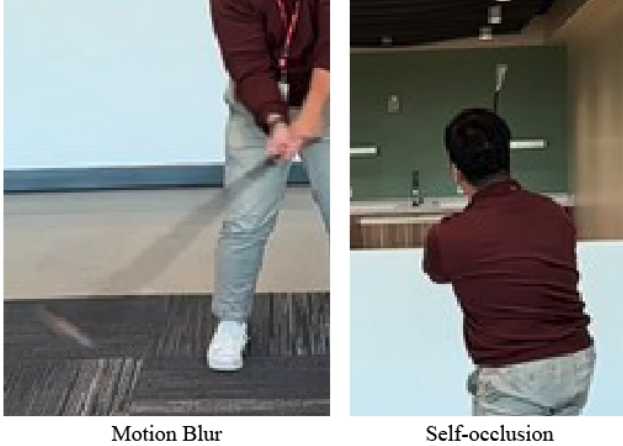


Figure 2. Motion blur and self-occlusion in golf swings. Motion blur (left) occurs due to fast swing speed producing multiple ghost images of the club. Self-occlusion (right) frequently occurs when the player’s body hides the shaft.

1. Introduction

Golf performance depends critically on precise postures of both the athlete and the club. At impact the club speed and orientation largely determine the ball’s direction, trajectory, and carry distance [33]. Recent computer vision studies have produced substantial work on human body pose for coaching and feedback in sports [2, 4, 11, 18, 37] and in golf specifically [10, 15, 21]. While these approaches improve swing consistency and recognize body posture errors, club-focused methods that target fine-grained club dynamics (Figure 1) remain largely overlooked. Accurate estimation of the club pose is essential for richer swing diagnostics.

Estimating club pose is technically challenging due to the club’s physical properties and fast motion. The club is a thin elongated and partly reflective object that undergoes rapid swing motion inducing severe motion blur especially around impact (Figure 2) [17, 32]. Self-occlusions by the hands and body are frequent throughout the swing. Club appearance also varies across types as drivers and woods have large heads while irons consist of thin metallic faces at various loft angles. Beyond these visual challenges practical deployment requires real-time inference to support live feedback during practice sessions.

Prior work in golf vision has predominantly focused on human body pose analysis [20, 27, 38]. A few studies consider the golf club alongside players [12, 19] but rely on markers with limited club types. Purely club-centric pose estimation in markerless settings remains underexplored. Existing methods also lack diversity in participants, club types, viewpoints, and environments needed for robust generalization. Furthermore, most pose estimation archi-

tectures prioritize accuracy without considering inference speed, which limits their applicability in real-time coaching systems.

In this paper, we tackle markerless club-only pose estimation with an emphasis on real-time inference. We collect a dedicated dataset across multiple subjects, environments, and club types with a universal five-keypoint schema that applies to all club geometries (Section 3.1). We propose CADDIE, a compact two-stage architecture that uses lightweight ConvNeXtV2-Atto [34] backbones in both detection and keypoint stages totaling 14.3M parameters to achieve 69 FPS while outperforming methods with $6\times$ more parameters (Section 3.2). We further introduce temporal smoothing that reduces detection frequency through adaptive interpolation to reach up to 238 FPS with minimal accuracy loss (Section 3.3).

Our main contributions are as follows:

- We present what is to our knowledge the first comprehensive study focused solely on markerless golf club pose estimation excluding human pose. The GolfClub dataset covers 7 subjects, 6 locations, and 3 club types with 72K annotated frames.
- We propose CADDIE, a lightweight two-stage architecture (9.5M detection + 4.8M pose = 14.3M total) that achieves 88.52% PCK@5px at 69 FPS, outperforming HRNet (28.6M), ViTPose (90.0M), and YOLO-based methods in both accuracy and speed.
- We introduce an adaptive temporal smoothing pipeline that scales inference from 69 to 238 FPS with less than 1% accuracy degradation through multi-level bbox interpolation.

2. Related Work

2.1. 2D Pose Estimation

Two-dimensional pose estimation follows two main paradigms. Top-down methods first detect instances with an object detector and then estimate keypoints within each cropped region [28, 35]. Bottom-up methods predict all keypoints jointly and group them through learned association cues [3, 7]. Top-down pipelines generally achieve higher accuracy because the cropped region removes background clutter and normalizes scale. HRNet [31] maintains high-resolution feature maps through parallel multi-scale branches and remains a strong top-down baseline. SimpleBaseline [35] demonstrates that a ResNet backbone followed by deconvolution layers can match more complex designs. More recently ViTPose [36] applies vision transformers to keypoint estimation and achieves state-of-the-art accuracy on COCO [22] at the cost of large model sizes exceeding 90M parameters. On the efficiency side, YOLO-Pose [26] and YOLO8-Pose [14] integrate keypoint heads into single-stage detectors for fast inference. These single-

stage approaches sacrifice localization precision because keypoints share the detection backbone without a dedicated high-resolution refinement stage.

Our method follows the top-down paradigm with two lightweight stages that each use a ConvNeXtV2-Atto backbone [34]. The detection stage contains 9.5M parameters and the pose stage contains 4.8M parameters for a total of 14.3M. Unlike HRNet and ViTPose that require large backbones for high accuracy CADDIE achieves competitive keypoint precision through compact feature pyramids and heatmap-based prediction while running at 69 FPS.

2.2. Lightweight Detection and Efficient Backbones

Real-time pose estimation demands an efficient object detector in the first stage. RetinaNet [24] introduces focal loss to address foreground-background class imbalance in dense single-stage detection. Feature Pyramid Networks (FPN) [23] fuse multi-scale features from backbone stages to handle objects at varying scales. Modern efficient backbones such as EfficientNet [29] and MobileNetV3 [8] reduce computation through depth-wise separable convolutions. ConvNeXtV2 [34] modernizes the ConvNet design with masked autoencoder pretraining and global response normalization to match transformer accuracy at lower parameter counts. The Atto variant provides a strong accuracy-to-parameter ratio with only 3.7M parameters making it suitable for latency-sensitive applications.

CADDIE pairs ConvNeXtV2-Atto with RetinaNet and FPN for Stage 1 detection completing club localization in 8.4 ms. We transfer classification and regression head weights from a COCO-pretrained ResNet50-FPN-v2 RetinaNet to compensate for the smaller backbone capacity. This weight transfer strategy accelerates convergence on limited domain-specific training data without increasing inference cost.

2.3. Sports Pose Estimation and Golf Swing Analysis

Pose estimation in sports addresses domain-specific challenges including high-speed motion and motion blur [17, 32] as well as frequent self-occlusion and cluttered backgrounds. Several works apply human body pose to coaching feedback systems for tennis [30] and general athletics [2, 11, 25]. Golf-specific studies have focused on human body kinematics for swing classification and coaching [10, 15, 20, 27, 38]. GolfPose [12, 19] incorporates club information alongside body pose but relies on physical markers and covers limited club types.

Equipment-centric pose estimation that targets the golf club rather than the human body remains largely unexplored. The club presents distinct challenges compared to human pose. It is a thin elongated object with minimal texture and its appearance varies substantially across drivers,

woods, and irons. Existing human pose methods are not designed for these geometric properties. Our work addresses this gap through a dedicated five-keypoint schema that captures the full club geometry across all club types and a compact architecture optimized for the unique visual characteristics of golf equipment.

2.4. Temporal Smoothing in Video Pose Estimation

Video-based pose estimation benefits from temporal consistency to reduce frame-to-frame jitter and handle transient detection failures. Kalman filtering [16] provides a classical approach to smooth trajectories under a constant-velocity motion model. Exponential moving average (EMA) filters offer a simpler alternative for stabilizing bounding box dimensions across frames. Recent deep learning approaches use temporal transformers [1, 40] or recurrent networks to enforce temporal coherence but add substantial computational overhead that conflicts with real-time requirements.

An orthogonal strategy reduces inference cost by running the detector on a subset of frames and interpolating detections for intermediate frames. Linear interpolation is the simplest option while higher-order methods such as quadratic and cubic spline interpolation capture acceleration patterns in non-linear motion. CADDIE combines Kalman filtering for box center smoothing with EMA for box dimension smoothing at the per-frame level. For higher throughput we introduce an adaptive multi-level interpolation scheme that selects among linear, quadratic, and Hermite interpolation based on motion characteristics. This temporal pipeline scales inference from 69 FPS to 238 FPS with less than 1% accuracy degradation.

3. Methodology

We propose CADDIE (Compact Adaptive Detection-Driven Inference for Real-Time Golf Club Pose Estimation) as a lightweight two-stage framework designed for real-time golf club keypoint estimation. Section 3.1 describes the GolfClub dataset. Section 3.2 presents the CADDIE architecture optimized for inference speed. Section 3.3 introduces temporal smoothing that further increases throughput by reducing detection frequency.

3.1. GolfClub Dataset

We collect the GolfClub dataset using multiple cameras (Baumer, two GoPro models (Hero 10 and Hero 13), and iPhone 14 Pro Max) without markers across six locations spanning indoor and outdoor environments (Figure 4). The dataset contains 67 clips from 7 subjects and additional license-free public videos from Pexels totaling 72,012 frames. The Pexels videos contain bounding box annotations only and are used exclusively for Stage 1 detection training. Each clip captures a golf swing with one of three club types: driver (42 clips), iron (13 clips), or wood

Table 1. GolfClub dataset distribution. The dataset spans 6 locations with indoor and outdoor environments. The training subject does not appear in validation or test sets. One subject appears in both the validation and test sets with non-overlapping clips, yielding 7 unique subjects in total.

Split	Clips	Frames	Subjects	Indoor	Outdoor
Train	43	55,792	1 + Pexels	15	28
Val	7	10,097	2 + Pexels	2	5
Test	17	6,123	5 + Pexels	16	1
Total	67	72,012	7* + Pexels	33	34

*One subject appears in both the validation and test sets with non-overlapping clips.

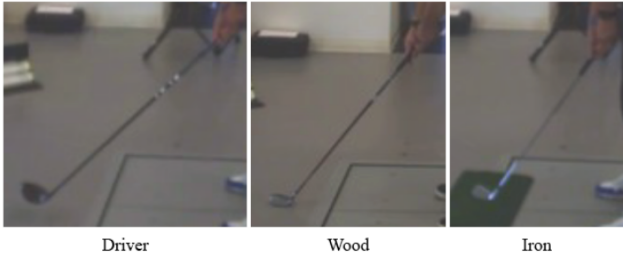


Figure 3. Club types in GolfClub: driver, wood, and iron. Drivers have the largest club heads and wide faces. Woods come with compact heads. Irons have thin metallic faces with various loft angles. Our universal keypoint schema applies consistently to all types.

(9 clips). Frames are captured at 200–300 FPS to support high-speed motion analysis where rapid club movement introduces severe motion blur (Figure 2).

We define a universal five-keypoint schema applicable to all club types: grip, shaft, neck, head-in, and head-out (Figure 5). The grip marks the top of the handle. The shaft is at the junction between grip and stick. The neck connects the shaft to the club head. Head-in and head-out represent the closest and farthest points of the club face from the player. This configuration captures the full club geometry and remains consistent across drivers, woods, and irons (Figure 3).

The dataset follows a subject-wise split to evaluate generalization to unseen subjects. The training set contains 43 clips (55,792 frames) from 1 subject across 6 locations plus public videos. The validation set has 7 clips (10,097 frames) from 2 subjects. The test set has 17 clips (6,123 frames) from 5 subjects with no overlap in subjects between splits. Table 1 provides the detailed distribution.

3.2. CADDIE Architecture

CADDIE is a compact two-stage framework designed to maximize inference speed while maintaining high keypoint accuracy (Figure 6). The full pipeline processes a 640×360



Figure 4. Two viewpoints at an outdoor scene. The *front* view (left) captures the full swing trajectory. The *head-face* view (right) is positioned toward the club face so it is clearly visible at impact.

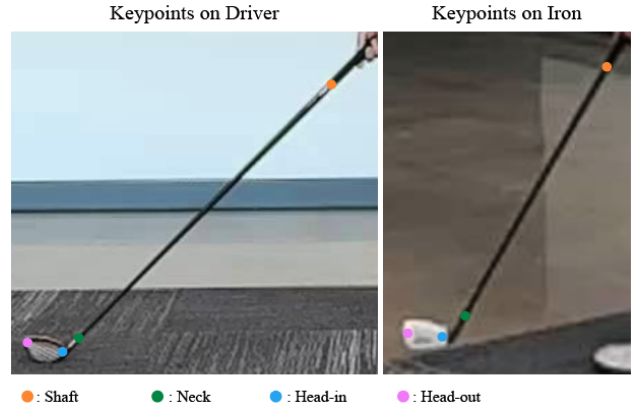


Figure 5. Universal keypoint schema for driver, wood, and iron. Five keypoints (grip, shaft, neck, head-in, head-out) are defined based on visual features and remain consistent across all club types.

image in 14.5 ms (69 FPS) on an RTX 3090 GPU. Both stages use ConvNeXtV2-Atto [34] backbones pretrained on ImageNet-1K [6]. Stage 1 (detection) contains 9.5M parameters including the RetinaNet head and FPN while Stage 2 (pose) contains 4.8M parameters for a combined total of 14.3M. This lightweight design is central to achieving real-time performance in a two-stage architecture where most competing methods require either much larger models or sacrifice the two-stage structure for speed.

3.2.1. Stage 1: Lightweight Detection

Stage 1 localizes the golf club in the full image using RetinaNet [24] with a ConvNeXtV2-Atto backbone. The backbone extracts multi-scale features from stages 2 and 3 which are fused through a Feature Pyramid Network (FPN) [23] with 256 channel output. This stage runs in 8.4 ms per im-

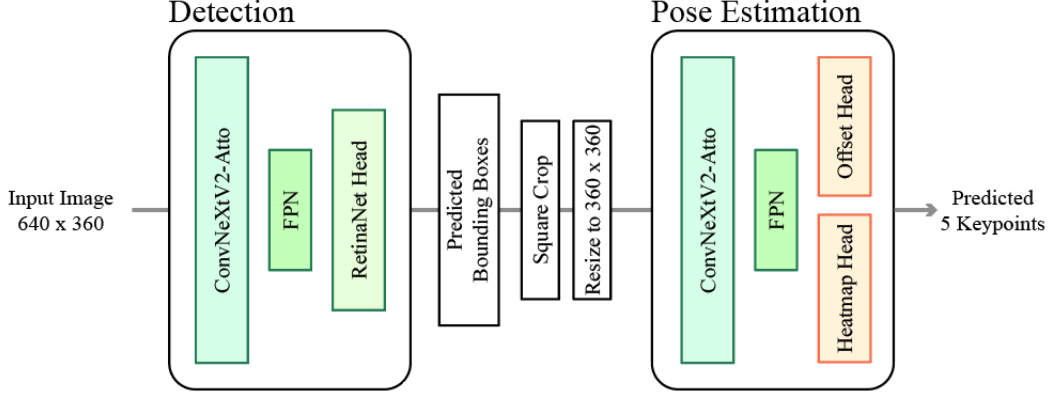


Figure 6. CADDIE pipeline. Stage 1 processes the full image (640×360) through ConvNeXtV2-Atto with FPN and RetinaNet to detect the golf club bounding box in 8.4 ms. The detection is expanded to a square crop (360×360) with padding. Stage 2 processes the crop through a second ConvNeXtV2-Atto with FPN and a refinement head to produce 5 heatmaps at 180×180 in 4.0 ms. Keypoints are extracted via argmax with offset refinement. The full pipeline runs at 69 FPS.

age.

To compensate for the small backbone we transfer detection knowledge from a pretrained COCO [22] ResNet50-FPN-v2 RetinaNet. The intermediate convolutional layers of the detection head are initialized from the COCO model while the final classification layer is randomly initialized for club-specific classes. This weight transfer accelerates convergence and improves detection quality on the limited training data.

The detection loss combines focal classification loss $\mathcal{L}_{cls}$ with Extended IoU (EIou) regression loss $\mathcal{L}_{EIou}$ [39]. EIou decouples width and height penalties to provide stable gradients for elongated objects:

$$\mathcal{L}_{EIou} = 1 - \text{IoU} + \frac{d^2}{c^2} + \frac{(\Delta w)^2}{w_c^2} + \frac{(\Delta h)^2}{h_c^2} \quad (1)$$

where d is the Euclidean distance between the center points of the predicted box and the ground-truth box, c is the diagonal length of the smallest box enclosing both the predicted and ground-truth boxes, and $\Delta w = w_p - w_g$, $\Delta h = h_p - h_g$ are the differences in width and height between the predicted box (w_p, h_p) and the ground-truth box (w_g, h_g), normalized by the width w_c and height h_c of the enclosing box.

We apply strong regularization to prevent overfitting: stochastic depth [9] at 0.4 drop rate, backbone stages 0–2 frozen, backbone learning rate at $0.05 \times$ base, weight decay of 0.08, and batch size of 8.

3.2.2. Stage 2: Heatmap-Based Keypoint Estimation

Stage 2 receives the 360×360 crop from Stage 1 and predicts five keypoint locations in 4.0 ms. A second ConvNeXtV2-Atto backbone with FPN (160-channel) extracts features at three scales. The finest FPN level passes through a refinement head with two convolutional blocks,

batch normalization, and $2 \times$ bilinear upsampling to produce 128-channel features at 180×180 .

Two parallel 1×1 heads operate on the refined features. The heatmap head produces $K=5$ probability maps through sigmoid activation:

$$\hat{\mathbf{H}}_k = \sigma(\mathbf{W}_h * \mathbf{F}), \quad k = 1, \dots, K \quad (2)$$

where $\mathbf{F}$ is the refined feature map and σ is the sigmoid function. The offset head produces $K \times 2$ displacement maps for sub-pixel refinement. Keypoint coordinates are extracted through argmax with offset correction:

$$\hat{\mathbf{p}}_k = \left(\arg \max_{(x,y)} \hat{\mathbf{H}}_k + \Delta \hat{\mathbf{p}}_k \right) \cdot \frac{s_c}{s_h} \quad (3)$$

where $s_c=360$ and $s_h=180$ give a scale factor of 2.

The Stage 2 loss combines MSE heatmap supervision over visible keypoints with L1 offset regression:

$$\mathcal{L}_{\text{Stage2}} = \lambda_h \mathcal{L}_{\text{hm}} + \lambda_o \mathcal{L}_{\text{off}} \quad (4)$$

where $\mathcal{L}_{\text{hm}}$ is computed against Gaussian targets ($\sigma=3.0$) and $\mathcal{L}_{\text{off}}$ penalizes sub-pixel offset error. We set $\lambda_h=1.0$ and $\lambda_o=0.1$ to prioritize coarse localization.

3.3. Temporal Smoothing for Increased Throughput

Beyond the 69 FPS baseline CADDIE can further increase throughput by reducing detection frequency and interpolating intermediate bounding boxes. This enables deployment at up to 238 FPS with less than 1% accuracy degradation.

3.3.1. Adaptive Multi-Level Interpolation

When detection runs on every N -th frame the system interpolates bounding boxes for intermediate frames through

three sequential levels. Linear interpolation fills gaps between detection anchors. A stationarity detector locks frames where the box center moves less than τ_s pixels to prevent drift during stationary phases such as the address position. Quadratic refinement is applied in sliding windows of three anchors with an acceleration check that falls back to cubic Hermite interpolation [5] when oscillation is detected. A spatial constraint blends toward linear when deviation exceeds τ_d pixels.

3.3.2. Kalman Filter Smoothing

For per-frame detection a constant-velocity Kalman filter smooths box centers with state $\mathbf{x} = [c_x, c_y, \dot{c}_x, \dot{c}_y]^T$. Box dimensions are smoothed separately via exponential moving average:

$$\hat{w}_t = \alpha \cdot w_t + (1 - \alpha) \cdot \hat{w}_{t-1} \quad (5)$$

where w_t is the detected width at frame t . This decoupled approach prevents coupling artifacts between position and size.

4. Results

We evaluate CADDIE against state-of-the-art pose estimation methods on the GolfClub dataset and report keypoint accuracy alongside inference speed. All experiments are conducted on a single NVIDIA RTX 3090 GPU with the test split of GolfClub (6123 frames). Section 4.1 presents baseline comparisons. Section 4.2 analyzes per-keypoint performance. Section 4.4 provides ablation studies on temporal smoothing.

4.1. Baseline Comparison

We compare CADDIE against five baseline configurations spanning three architectural families: YOLO direct pose estimation, YOLO two-stage detection-then-pose, HRNet, and ViTPose. Table 2 reports PCK at 5 and 10 pixel thresholds in 360×360 GT-crop space, OKS mAP at multiple IoU thresholds, mean pixel error, inference speed in FPS, and model size.

YOLO direct. YOLO26m-pose [13] and YOLO8m-pose [14] are trained from scratch on 640×360 images to detect and estimate keypoints in a single pass. Both use medium backbones (21.5M and 25.9M) but the single-stage approach struggles with fine-grained localization on thin elongated objects and achieves PCK@5px below 30%.

YOLO two-stage. YOLO26m followed by YOLO26m-pose on cropped regions improves accuracy to 73.03% PCK@5px by letting the pose model focus on the club region. The combined model has 41.9M parameters and runs at 60 FPS.

HRNet and ViTPose. HRNet-W32 [31] (28.6M) achieves 72.90% PCK@5px but runs at only 2.8 FPS. ViTPose-Base [36] (90.0M) with GT bounding boxes

reaches only 42.17% PCK@5px indicating that pretrained ViT features do not transfer effectively to golf club keypoints.

CADDIE (ours). CADDIE achieves 88.52% PCK@5px and 98.46% OKS mAP@0.50 at 69 FPS with 14.3M total parameters (9.5M detection + 4.8M pose). The heatmap-based Stage 2 provides dense spatial supervision critical for thin structures reducing mean error to 3.87 px compared to 5.28 px for the next best method (Figure 7).

4.2. Per-Keypoint Analysis

Table 3 reports per-keypoint performance for CADDIE and the baseline methods. CADDIE achieves the lowest mean error across all five keypoints in full-image pixel space.

The grip is the most challenging keypoint across all methods due to frequent occlusion by the golfer’s hands. CADDIE handles this with 93.84% PCK@5px and 1.06 px mean error while HRNet achieves only 53.84% and ViTPose drops to 7.36%. The head-out keypoint is second most challenging under motion blur where CADDIE achieves 81.42% PCK@5px still outperforming all baselines. The YOLO two-stage pipeline shows comparable performance on shaft and neck but produces no valid grip predictions as the grip falls outside its detection crop.

4.3. Inference Speed

Among methods with PCK@5px above 70% CADDIE is the fastest at 69 FPS (14.49 ms per image) with 14.3M parameters. The YOLO two-stage pipeline runs at 60 FPS but requires 41.9M parameters. HRNet runs at only 2.8 FPS. The YOLO direct models achieve higher FPS (120 and 154) but their keypoint accuracy is insufficient for swing analysis. CADDIE provides the best accuracy-to-speed trade-off through lightweight ConvNeXtV2-Atto backbones in both stages.

4.4. Ablation Study: Temporal Smoothing

We evaluate the impact of temporal smoothing on keypoint accuracy across video sequences. The ablation varies two axes: smoothing method and temporal stride (predict every N frames). Table 5 reports results for smoothing methods applied to every frame (predict every 1). Table 4 reports interpolation methods with temporal strides of 5, 10, and 20 frames.

4.4.1. Frame-by-Frame Smoothing

Table 5 compares four smoothing configurations applied to Stage 1 detections on every frame. All variants maintain PCK@5px within 0.1% of the baseline (88.52%). The Kalman+EMA combination slightly reduces mean error from 3.87 to 3.73 px while improving OKS mAP@0.50 from 98.46% to 98.53%. The smoothing overhead is negligible as it adds less than 1 ms per frame. This indicates

Table 2. Baseline comparison on the GolfClub test set (6123 frames). PCK and mean error are measured in 360×360 GT-crop space with padding ratio 1.3. OKS uses uniform $\sigma=0.1$ and object scale 360. † indicates GT bounding boxes are used. Best results are in **bold**.

Method	Params	FPS	PCK@5px	PCK@10px	Mean Err	OKS mAP@.50	OKS mAP@.90
YOLO26m-pose [13]	21.5M	120.4	29.25%	62.10%	26.29 px	84.80%	65.35%
YOLO8m-pose [14]	25.9M	153.7	24.02%	50.40%	45.79 px	78.44%	61.08%
YOLO two-stage	41.9M	60.0	73.03%	94.20%	5.28 px	93.71%	90.97%
HRNet-W32 [31]	28.6M	2.8	72.90%	78.42%	19.55 px	82.08%	61.44%
ViTPose-B† [36]	90.0M	47.3	42.17%	73.02%	16.34 px	79.14%	67.01%
CADDIE (ours)	14.3M	69.0	88.52%	97.11%	3.87 px	98.46%	97.17%

Table 3. Per-keypoint mean error (full-image pixels) on the GolfClub test set. Lower is better. CADDIE achieves the lowest error on all five keypoints. The grip column for YOLO two-stage is unavailable as the grip falls outside the detection crop in most frames.

	Grip	Shaft	Neck	Head-in	Head-out
YOLO26m	6.62	11.75	15.82	5.57	12.85
YOLO8m	6.98	12.35	33.44	7.71	36.35
YOLO two-stage	n/a	2.25	2.84	1.47	2.52
HRNet-W32	13.59	3.75	16.38	29.13	38.37
ViTPose-B	6.68	3.09	8.87	7.42	19.39
CADDIE	1.06	1.83	1.78	1.50	1.82

that the Stage 1 detections are already temporally stable and frame-by-frame smoothing provides marginal gains.

4.4.2. Interpolation with Temporal Stride

Table 4 reports results when Stage 1 detection runs on every N -th frame and intermediate frames use interpolated bounding boxes. This simulates a deployment scenario where reducing detection frequency increases throughput. At predict every 5 frames, the effective FPS increases from 82 to 185 while PCK@5px is maintained at 88.55% with the adaptive method. Linear interpolation performs comparably to more complex methods at this stride. At predict every 10 frames, FPS reaches 218 with less than 0.3% PCK degradation. At predict every 20 frames, the degradation is approximately 0.7% PCK which remains above 87.6% for all methods.

The adaptive method achieves the highest PCK@5px (88.55%) at predict every 5. Stationarity detection contributes 0.07% PCK improvement by preventing jitter during the address phase. The Hermite fallback shows its value at larger strides where it achieves 87.97% versus 87.78% for quadratic at predict every 20 by preventing oscillation artifacts.

4.4.3. Speed-Accuracy Trade-off

The temporal stride ablation reveals a favorable speed-accuracy trade-off. At predict every 5, the system achieves 185 FPS with no accuracy loss compared to the 82 FPS

baseline. At predict every 10, FPS increases to 218 with 0.27% PCK degradation. At predict every 20, FPS reaches 238 with 0.85% degradation. This demonstrates that CADDIE can scale from real-time (69 FPS with full detection) to high-throughput (238 FPS with temporal interpolation) depending on the application requirements. The linear interpolation method is sufficient for most practical scenarios as it matches or exceeds more complex methods while adding negligible computational overhead (less than 7 ms total across all frames).

5. Limitations

The current study has limitations that motivate future work. The dataset captures single-golfer single-club scenarios and does not address multi-person settings or simultaneous tracking of multiple clubs. Extending the framework to 3D club pose estimation through multi-view triangulation or monocular depth lifting would provide richer swing diagnostics. The temporal smoothing operates on bounding boxes and a natural extension is to apply similar interpolation directly to keypoint trajectories. Beyond golf the lightweight two-stage design and universal keypoint schema can be adapted to other sports equipment such as baseball bats and tennis rackets where similar challenges of thin elongated objects under fast motion arise.

6. Conclusion

We presented CADDIE, a compact two-stage framework that addresses real-time inference, markerless operation, and generalization across club types for golf club pose estimation. We introduced the GolfClub dataset with 72K annotated frames spanning 7 subjects, 6 locations, and 3 club types under a universal five-keypoint schema that captures the full club geometry from grip to head across drivers, woods, and irons. The subject-wise evaluation split ensures that all reported metrics reflect generalization to unseen individuals rather than memorization of training subjects. CADDIE achieves 88.52% PCK@5px at 69 FPS with only 14.3M total parameters (9.5M detection + 4.8M pose) by pairing lightweight ConvNeXtV2-Atto backbones

Table 4. Ablation on interpolation methods with temporal stride. Detection Frequency (DF) indicates how often Stage 1 detection runs (every N frames). Intermediate frames use interpolated bounding boxes. Higher FPS is achieved by reducing detection frequency. The adaptive method combines linear baseline with stationarity detection, quadratic refinement with Hermite fallback, and spatial constraints.

Method	DF	PCK@5px	PCK@10px	Mean Err	OKS@.50	OKS@.75	FPS
Linear	5	88.52%	97.16%	3.80 px	98.60%	97.93%	184.6
Quadratic	5	88.52%	97.13%	3.89 px	98.55%	97.97%	185.6
Cubic spline	5	88.40%	97.08%	3.96 px	98.51%	97.89%	185.1
Hermite	5	88.44%	97.11%	3.86 px	98.51%	97.93%	184.9
Adaptive (full)	5	88.55%	97.08%	3.90 px	98.55%	97.88%	184.9
w/o stationarity	5	88.48%	97.11%	3.89 px	98.55%	97.89%	183.8
w/o spatial	5	88.55%	97.09%	3.89 px	98.56%	97.89%	184.4
w/o Hermite	5	88.55%	97.08%	3.96 px	98.50%	97.83%	183.8
Linear	10	88.25%	97.13%	3.69 px	98.50%	97.88%	217.9
Quadratic	10	88.41%	96.99%	4.32 px	98.27%	97.68%	217.5
Cubic spline	10	88.52%	96.95%	4.32 px	98.32%	97.63%	217.6
Hermite	10	88.33%	97.12%	4.02 px	98.35%	97.70%	217.5
Adaptive (full)	10	88.29%	97.05%	4.04 px	98.38%	97.75%	217.0
Linear	20	88.14%	97.02%	4.38 px	98.11%	97.32%	237.9
Quadratic	20	87.78%	96.84%	4.64 px	98.20%	97.44%	236.6
Cubic spline	20	87.81%	96.92%	4.80 px	98.01%	97.14%	236.9
Hermite	20	87.97%	97.11%	4.44 px	98.22%	97.45%	237.9
Adaptive (full)	20	87.67%	96.89%	4.55 px	98.14%	97.13%	237.5
w/o stationarity	20	87.83%	96.94%	4.55 px	98.14%	97.16%	237.4
w/o spatial	20	87.75%	96.98%	4.72 px	98.27%	97.48%	237.5
w/o Hermite	20	87.67%	96.89%	4.55 px	98.14%	97.13%	238.5

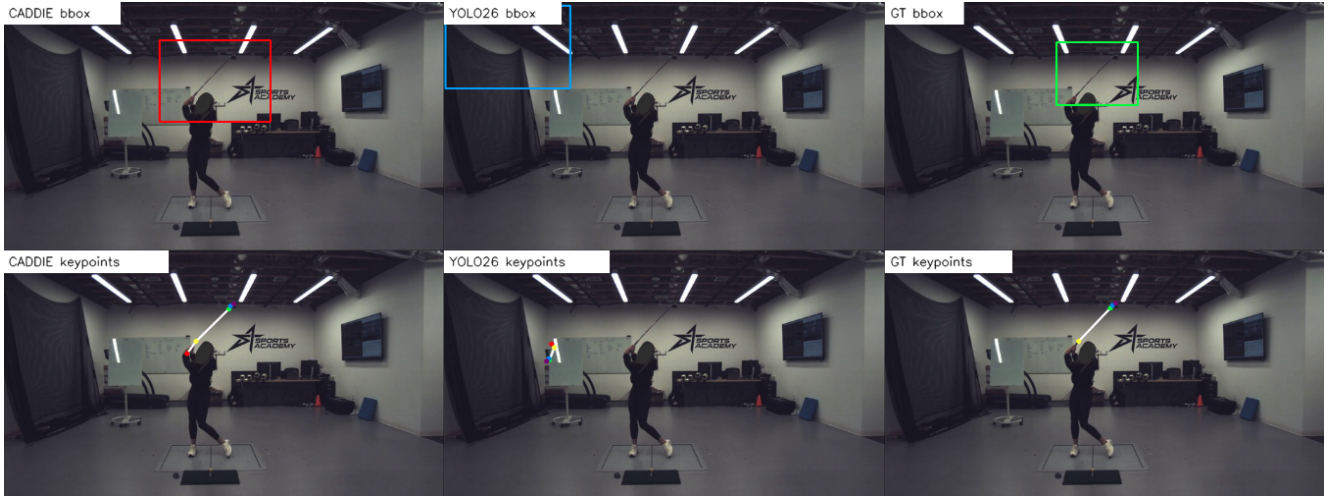


Figure 7. Qualitative comparison between CADDIE and two-stage YOLO26 in a dim indoor environment with background clutter. Top row shows predicted bounding boxes and bottom row shows predicted keypoints alongside ground truth. YOLO26 produces an oversized bounding box that includes background objects and fails to localize keypoints on the club. CADDIE produces a bounding box close to the ground truth and accurately estimates all five keypoints along the club shaft and head despite low lighting and visual distractors in the background.

with RetinaNet detection and heatmap-based keypoint regression. This outperforms HRNet (28.6M) by 15.6% PCK@5px, ViTPose (90.0M) by 46.4% PCK@5px, and the two-stage YOLO pipeline (41.9M) by 15.5% PCK@5px

while maintaining real-time throughput suitable for live coaching scenarios. The adaptive temporal smoothing pipeline further scales throughput from 69 FPS to 238 FPS with less than 1% accuracy degradation through multi-

Table 5. Ablation on frame-by-frame smoothing methods (predict every 1 frame). All variants detect every frame and apply post-hoc smoothing to bounding boxes before Stage 2. The baseline (none) uses raw detections.

Method	PCK@5px	Mean Err	OKS@.50	FPS
None (baseline)	88.52%	3.87 px	98.46%	81.9
Kalman	88.52%	3.99 px	98.42%	83.2
EMA	88.43%	3.87 px	98.46%	83.4
Kalman+EMA	88.50%	3.73 px	98.53%	83.1

level bounding box interpolation combining stationarity detection, quadratic refinement, and Hermite fallback. Per-keypoint analysis confirms that CADDIE maintains low error across all five keypoints including the challenging grip under frequent hand occlusion. These results demonstrate that accurate and robust golf club pose estimation is achievable in real time with compact models making CADDIE practical for deployment in live swing analysis and coaching feedback systems.

Acknowledgements

This work was conducted during a research internship at Fujitsu Research of America. The authors thank the team at Fujitsu Research of America for their support and valuable discussions throughout this project.

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